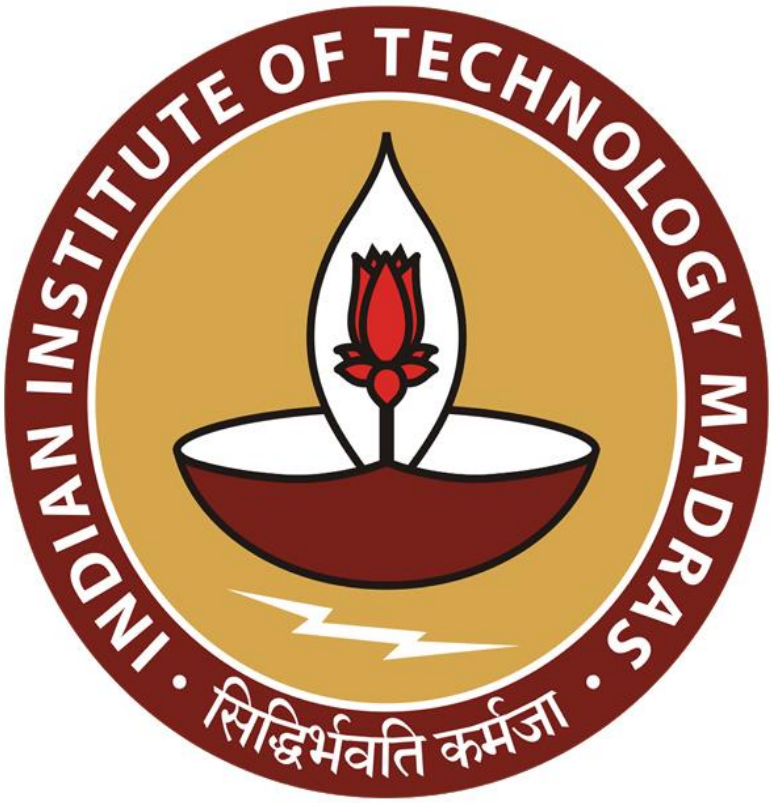
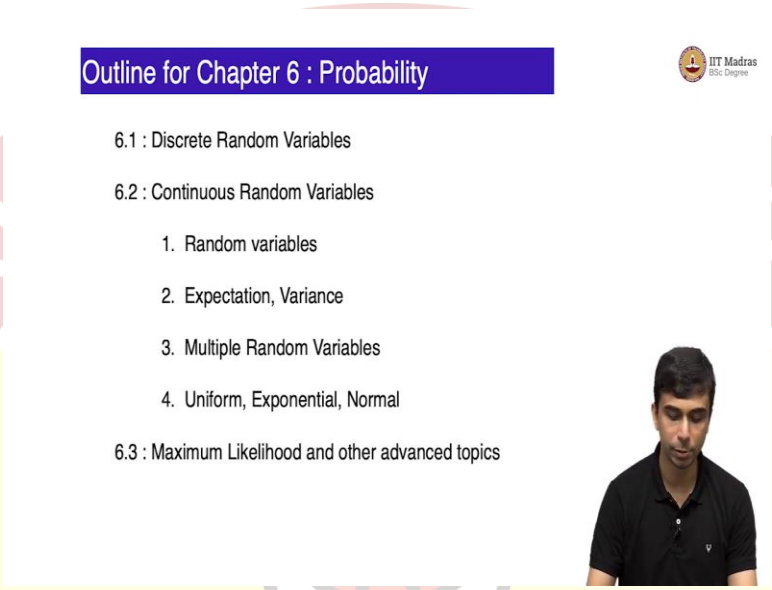


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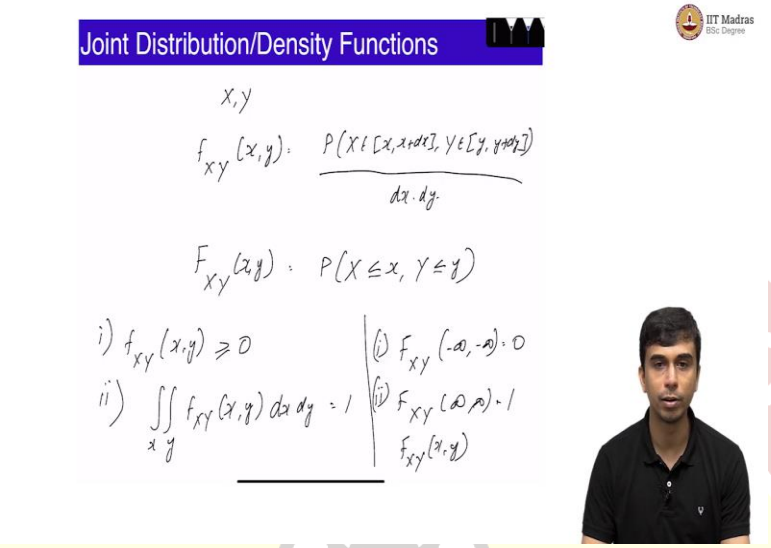
Professor Harish Guruprasad Rmaswamy
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Multiple Random Variables

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Once again until now you have been dealing only with single random variables. Their properties such as density functions, cumulative distribution functions, expectation, variance, and so on. Now, we will deal with multiple random variables, the same way we will see very similar concepts. For us we had for discrete random variables but just that how would those extend for continuous random variables in the multiple random variable setting we will see now.

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The first thing is to generalize the if you have a random variable x and y, the same experiment of course. The joint density function or joint, we will call the joint density function is typically denoted by $f_{XY}[x, y]$ is well, its probability of x belonging to (x + dx) and y belonging to [y to y + dy] divided by × dy. So, you can see it is a density because it is strong 2 dimensional you have to divide by the area instead of by the length.

So, for the one dimensional for just one single random variable you just have probably have x belonging to [x + dx] divided by dx. But now we have two random variables, the joint density function is simply the probably the x , y belongs to this rectangle given by x ,x + dx y and y + dy divided by the area of the rectangle which is dx dy. So, that is why it is called a density function.

And you can clearly see how this would extend for let us say 3 random variables or 4 random variables only thing that changes you would have let us say if you had 3 random variables it would be x belonging to x + dx y belong to y + dy and z belonging to z + dz divided by dx into dy into dz. So, that would also be a density function. But 2 random variables is the easiest, easy to write and you can easily see how it generalizes for higher random variables also.

We will only work with 2 random variables but there is nothing stopping you from working with more than 2. That is the joint density function, you can also have the joint cumulative distribution function which is denoted by capital F, capital F_{XY} (x , y) is equal to probability of P(X ≤ x and

Y ≤ y). So, this is exactly the same as we had for discrete random variables there is no not even any dx dy.

Because cumulative distribution functions are probabilities, there is no densities there. So, it is looks exactly the same. So, what are the properties? The properties, first of all these are still probability. So, F_{XY} (x , y) is should still be greater than 0. But it can be greater than 1 also. But the integral for single random variables you have the integration to 1, the area under the curve was equal to 1.

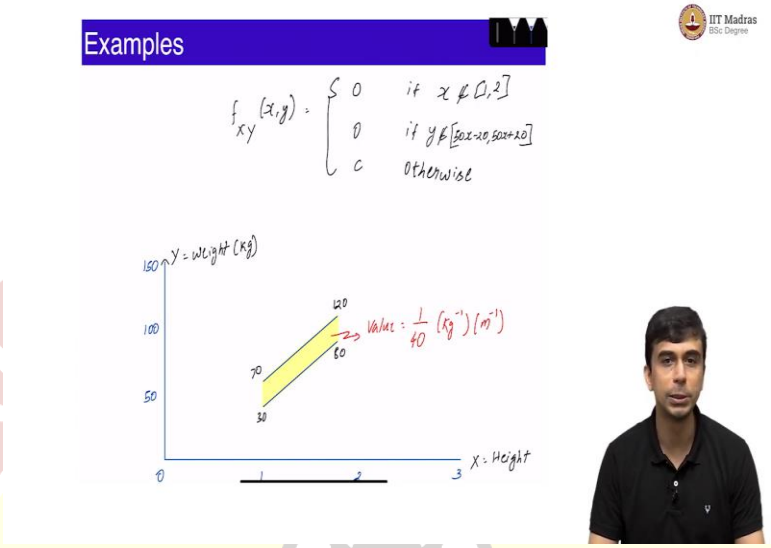
For let say for 2 random variable case you what you will have is the volume under the surface given by F_{XY} is equal to 1. What does that mean? Well, the some properties, we will quickly write F_{XY} of x, y is greater than or equal to 0 and integral over x the lower y F_{XY} (x , y) is equal to 1. I forgot the dx dy.

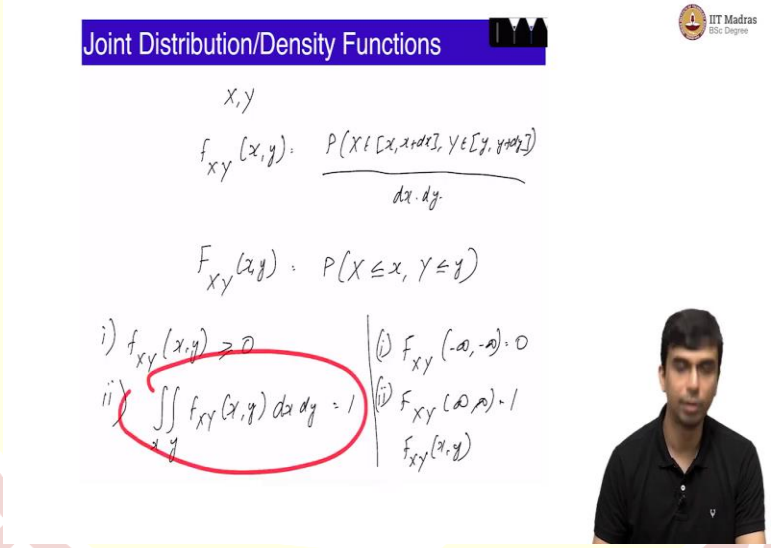
These are the simple generalizations from single random variable to multiple random variables. Similarly, you have the same CDF thing which is $F_{XY}(-\infty, -\infty) = 0$ F_{XY} $(\infty, \infty) = 1$. So, there are some extra properties which come as a result of it being a function of 2 variables. But we will see that pretty soon. So, these are basic properties.

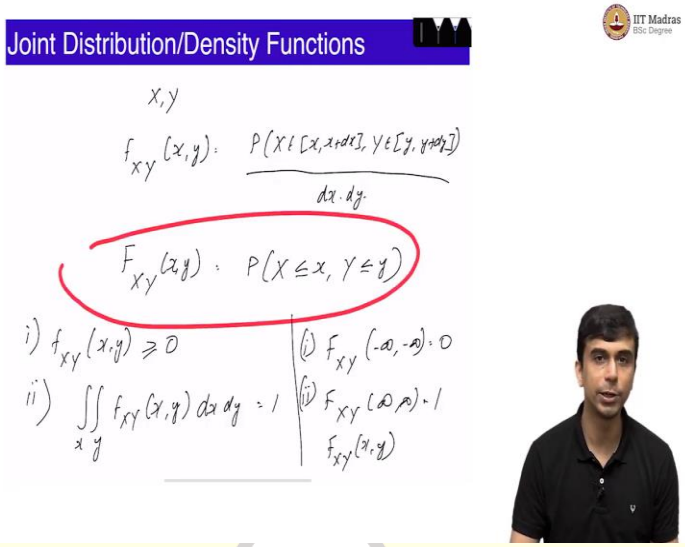
And once again, you have that F_{XY} of by the cumulative by the monotonic nature, $F_{XY}(x,y)$ is monotone in both x and y. By that what do I mean that F_{XY} of 3, 5 is definitely going to be greater than or equal to F_{XY} of let us say 2.9, 5. And similarly, F_{XY} of 3, 5.1 is definitely going to be greater than or equal to F x of 3 , 5 as long as one of the values is even higher the probability is this is strictly larger probability or strictly not less or in it can be equal but it can never be less.

Probability of x ≤ 3 and y ≤ 5 is strict is at least as big as probability of x ≤ 3 and y ≤ 4. So, 3 even if one of the values y increases the CDF has to either stay the same or increase it cannot decrease. So, that is why F x y is monotone in monotone increasing in both its arguments both x and y. So, that is the joint.

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So, now let us give a very simple example. Let us say you pick a person at random and you measure their height and weight. So, that is the experiment. And the random variable X is let us say, let us call x to be the height and y to be the weight of the person. And let us say the density function happens to be this $F_{XY}(x,y) = 0$ if the height which is x does not lie between let us say 1 meter to 2 meters.

So, people are uniform. I mean no there is no one below and there is no one greater than 2 meters. It is a very simple example. Keep better that in mind. That is your height. For a given height though what is the weight range? So, it is also equal to 0 if y does not belong to 50x - 20 and 50x + 20. By that what do I mean? I mean that if x is 1 meter the height ranges between 30 to 70, if x is 2 meters the height range is, sorry, the weight range is between 80 to 120.

So, Fx was 1 meter the height + 1 meter the weight range is between 30 kilograms to 70 kilograms. If height was 2 meters the height range is between 80 kilograms to 120 kilograms. And if x, y are the range of if x is in the range of 1 to 20, sorry, 1 to 2 and y is in the range of 50x - 20 to 50x + 20 then we have let us say some constant C otherwise. What is this constant going to be? We do not mean, we do not know now, but what can you use to get this?

We can use the property of the density function which is that if you integrate it you get 1. So, if you integrate the density function over the range of the random variables you get 1. So, let us do a simple visualization of this distribution. Let us say x axis is the height and y axis is the weight you

can see that x axis 0, 1, 2, 3 and Y axis 50, 100, 150. We can see that only in the range of 1 to 2 there is something interesting happening for height.

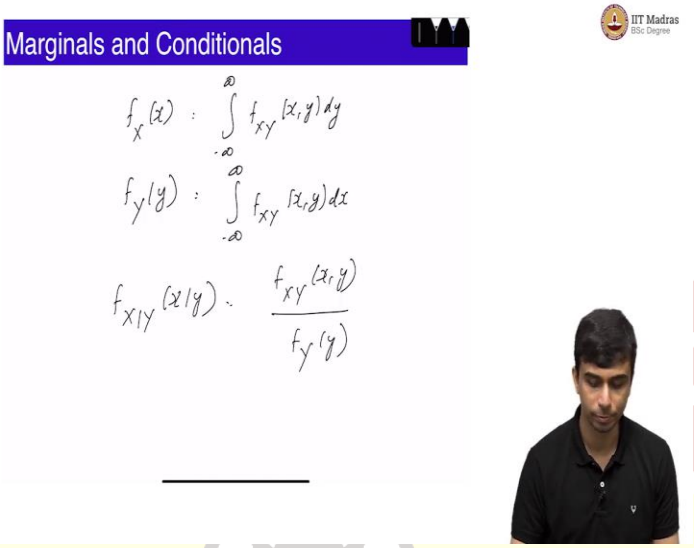
And we will within the range of 50x - 20 and 50x + 20. So, this is 30. This 70, this is 80, this is 120. So, only within, only if x, y lies within this parallelogram you have $F_{XY}(x,y)$ is not 0 which is equal to constant some constants are outside that it is going to be equal to 0. But how will you find the value of this constant? Well, you can just find the volume under this surface. Because it is a flat I think it is simply exactly equal to the area under area of the shaded region.

And the area of the shaded region is well it is 40-kilogram meter. So, the constant should be 1 by 40 per kilogram per meter. So, this constant actually can take it to have units which is kilograms or - 1 meter or - 1. So, this C would be equal to 1 by 40. If the unit you are using is kilogram and meters. This is a simple example of multiple random variables having a joint distribution, you can clearly see that x and y are not independent.

But if not define what independence is I mean in an English sense you can see x and y are dependent. Based on the way we have defined the function. We will make a formal definition of independent random variables pretty soon. But in the English sense you can see they are they x and the height and weight do seem to depend on each other.

So, this is an example of a joint density function you can use the simple logic of the cumulative distribution function to get the cumulative joint cumulative distribution function also. But that is not a very interesting object for now. Let us skip ahead.

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Now, we can define now we define the joint distributions and densities. We can also define marginal distributions and conditional distributions in the same way that we did for discrete random variables. What is the marginal density of x and y random variables? What is F x of x from $F_{XY}(x,y)$ can you get $F_X(x)$. In fact, it can $f_X(x)$ is simply equal to if you integrate over y integral $(-\infty, \infty)$. $f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,y)dy$

$F_{XY}(x,y)$ dy you get your distribution of your density of x. You can see this is exactly analogous to the discrete case where you your row sums correspond to a marginal and column sums also correspond to a marginal. Similarly, you can also get your density of y the marginal density of y from the joint density of x and y by integrating out x $f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x,y)dx$.

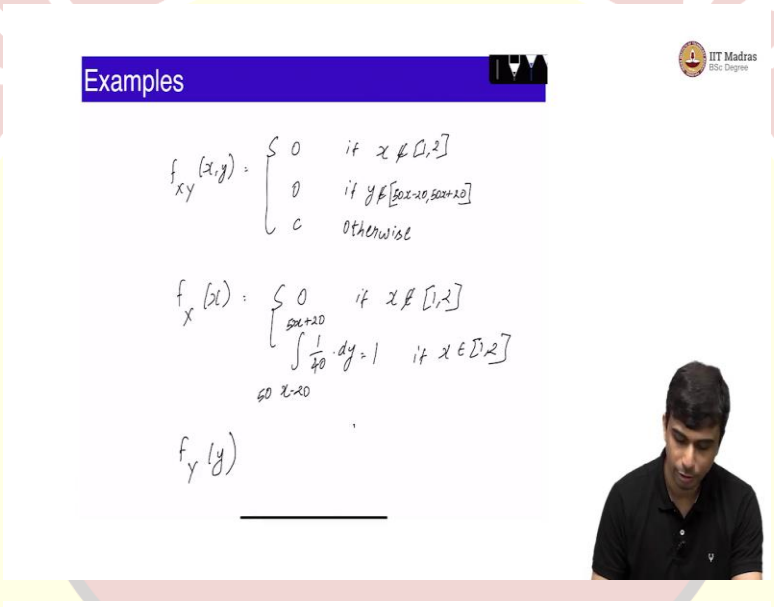
You can get back your marginal densities from the joint density simply by integrating out the other variable. You can also define conditional densities now. What is that? So, $\frac{f_{XY}(x,y)}{f_Y(y)}$ So, this technically makes sense only when $f_Y(y)$ not 0. If $f_Y(y)$ is 0 then Fx a given y of x given y does not make sense.

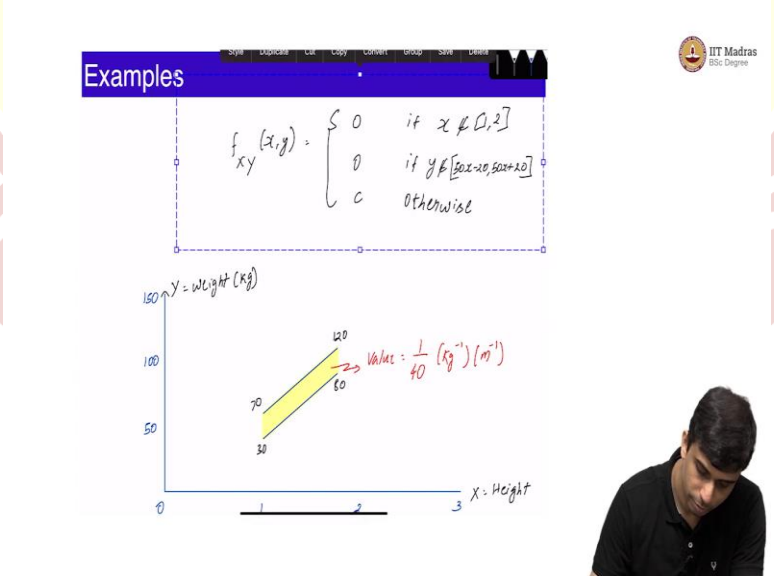
So, this will only will define this only the condition will be defined only for the cases where the marginal Fy is not 0. So, we will give an interpretation the interpretation is pretty simple, the same way that you had for a discrete case Fx the marginal are simply the density functions of the

individual random variables the conditionals are given for example given the height is 1 meter, what is the distribution of weight?

Given the height is someone let us say 1.7 meters? What is the distribution of weight are given the weight is let say 75 kilograms? What is the distribution of height? What is the density of height? You can ask these questions. And the right tool for answering these questions is the conditional densities or conditional distributions. So, let us evaluate all of these for the example that we had.

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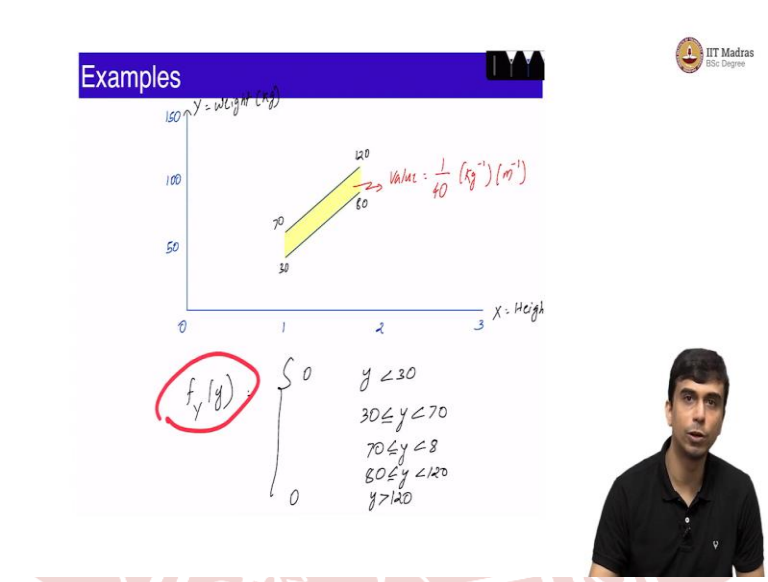
Let us recall the example again. Let us actually copy paste that let us not. Or let us come back to this place here. And let us take that same example. So, what is $f_X(x)$ going to be? x is the height

and y is the weight. So, $f_X(x)$ it is clearly equal to 0 if x does not belong to 1 , 2. But within the range of 1 to 2, what is that? Well, that is $\int_{-\infty}^{\infty} c\, dy$. What is C? C is $\frac{1}{40}$ times dy.

And what is instead of - infinity to infinity you know that it is actually nonzero only within the range of (x – 20) to (x + 20). You can just use this by saying (x - 20) to (x + sorry 5x 50x - 20 to (50x + 20). And this is exactly equal to 1. So, the marginal density of x is equal to is simply uniform in the range of 1 to 2. And that does make sense where if you can see from the graph here that within the range of 1 to 2 the width seemed uniform.

So, the probability of the random variable random variable x is simply uniform with range of 1 to 2. y on the other hand is slightly more complicated. You can see that the width is for a given y some y's are not. Not all y's is with nonzero density or equally likely. In fact, a y of 30 is less likely than something like let us say 75. Why is that? Well, we will get the density function of y also. What is a $f_Y(y)$? Here you have to use different cases now.

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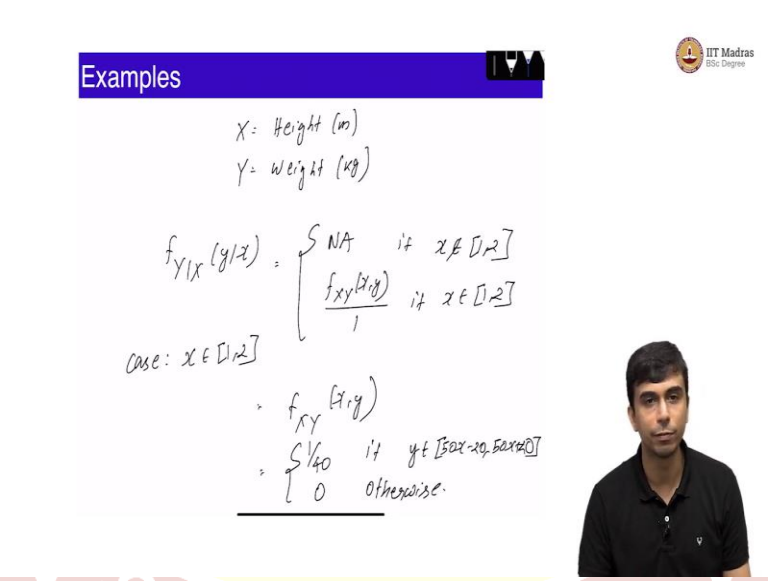
Let us actually copy paste that and get the answer. Now for $f_Y(y)$ let us use a different limit, $f_Y(y)$ we will just first get the cases out first. For what is the weight? It is going to be 0 if y is less than 30. And there is going to be other interesting is for y between the range of 30 to 70 and 70, to 80 and finally, 80 to 120 and y greater than 120. So, these are the 5 cases that you can think of.

And immediately the 2 cases that is why less than 13y is greater than 120 are going to be 0. You only have to consider these 3 cases. So, for each of these cases, for let us say for y in the range of 30 to 70 you are going to be integrating across x from 1 to where so, that is going to be the case. For y in the range of 70 to 80. We will be integrating across some other limits. So, from based on the value of y you have to get the limits of x. And that will decide the density of y.

So, this is going to be a simple exercise for you get the density of y from the joint density of x and y. So, it is a marginal guy is a pretty simple exercise. And you should also check that a simple sanity check would be $f_Y(y)$ if you integrate from -∞ to ∞ it should be equal to 1 because. This is also a valid density function.

Now, I think we also need to do some conditional distributions. But we will do some conditional distributions. So, continuous distributions are pretty easy to do. Let us do that.

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Let us take the same example x is the height and y is the weight m meters' y is the weight in kilograms. What is =Fx given y of x given y or let us do the y given x case first just easier to think of. Recall a had set that this function makes sense only use $f_X(x)$ is not 0. Which means that this entire thing makes sense only when x in the range of 1 to 2. Why is that? Why is that not a restriction because outside the range of 1 to 2x is never going to happen.

So, you do not need to worry about such events. So, $f_{Y|X}(y|x)$ we are only going to define it in the range of for x in the range of 1 to 2. If x is outside the range of 1 to 2 I am not even going to define

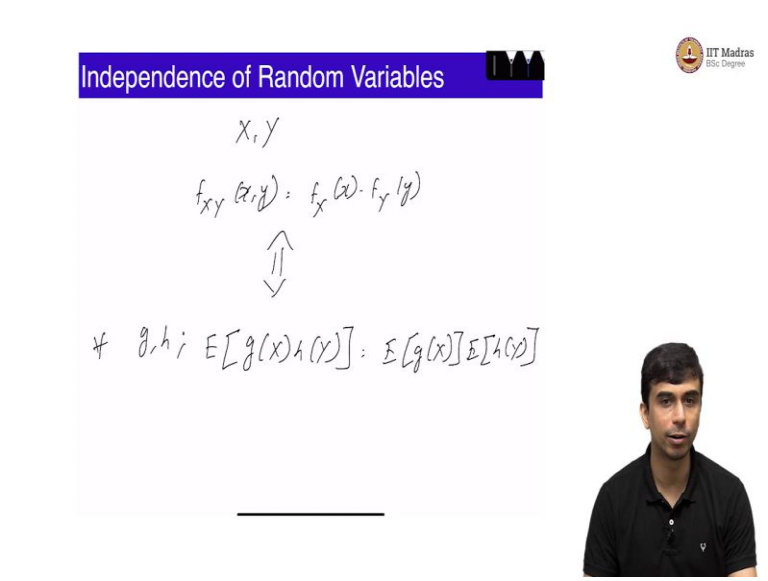
it, we can define it however you want. In fact, not defined in a if x does not belong to 1 to 2 within the range of 1 to 2.

We can define$F_{Y|X}(y|x)$. if x does belong to 1 to 2 F y given $f_X(y)$ given x is simply F_{XY}(x,y) divided by F y of sorry, $f_X(x)$ and $f_X(x)$ we happen to know x is uniform in the range of 1 to 2 that is equal to 1. So, F_{XY} (x, y) if x belongs to. So, in other words Fx Fy given $f_X(y|x)$ is equal to F_{XY}(x,y) if x belongs to, I am not going to bother about the case where x does not belong to 1 , x belongs to 1 , 2.

That is the F_{XY}(x,y) case x belongs to 1,2. So, which is equal to 1 by 40 if y belongs to 50x - 20 to 50x + 40 0 otherwise. So, this is the conditional density function of y given x. You can do the same thing for x given y also be slight bit more intricate but the logic is the same.

So, by this we can say that the continuous distribution of y given x is uniform in the range of (50x – 20) to (50x + 20), where I write 40, 20. So, this with this we can wrap up the basic idea of conditional densities and distributions and marginal, in short, will often just call it conditionals and marginal. The word density or distribution will indicate whether it is a density or a cumulative distribution function.

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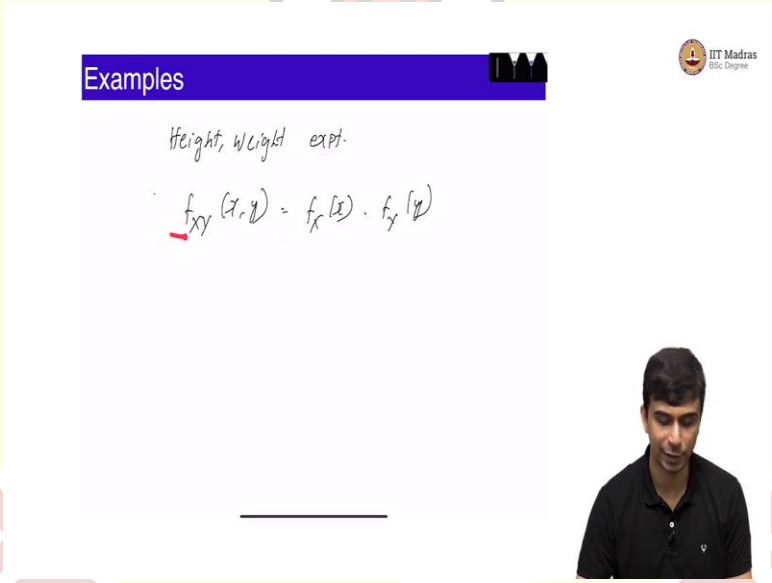
Finally, we move on to the next concept which is independence of random variables. We have defined independence of random variables for discrete random variables before the same. Well, a very similar idea applies to independence of continuous random variables also. Let us say you

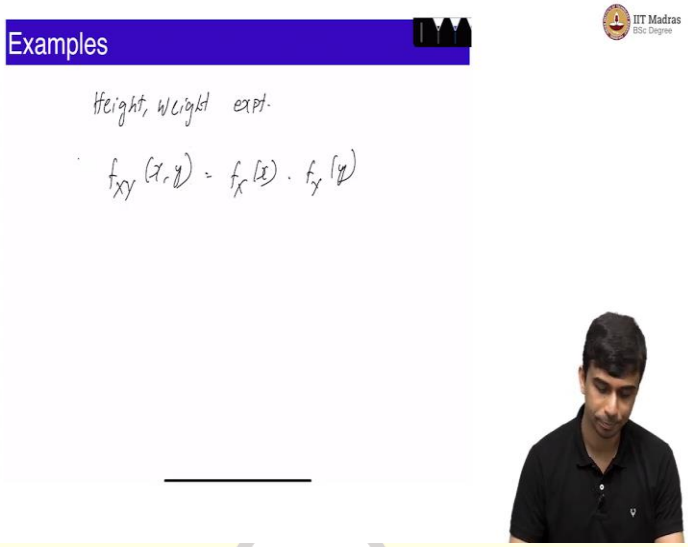
have a random variable X and Y. We say these two random variables X and Y are independent. If $F_{XY}(x,y)$ is equal to $f_X(x) \times f_Y(y)$

It is as simple as that in the same way we had four discrete random variables the definition is exactly the same. There are several equivalent definitions which is for example the equivalent definition is that you have 2 functions let us say any two functions g, h. For any 2 functions for all functions g, h expected value of g(x) into h (y) is equal to expected value of g (x) into expected value of h(y).

This is another equivalent definition for independence of random variables. We have seen how we have seen how to prove this for the case of discrete random variables you can do the same thing for continuous random variables also.

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So, let us consider the example, the concept, the height weight example. In that case do we have $F_{XY}(x,y)$ for the height weight experiment. Do we have this $F_{XY}(x, y)$ equals F y of y do we have this. We can easily see it is not the case. Why is that? For example,? Very easy check is to look for 0s. So, you can see that here. We have $f_X(x)$ y the range of let us say this particular square let us say, very small square here.

Let us say for example say 70, 1.9 are I think that is a reasonable $F_{XY}(x, y)$ of 70 1.9. The density is 0, because 70 is below 50 into 1.9 - 20. So, it is 0. So, the only way is that if you have F(x ,y) of 1.9 is 0. It can be independent only Fx of 70 is 0 or Fx of Fy of sorry Fx of 1.9 is 0 or F y of 70 is 0. But we know neither of that is true. So, in this case, we have F_{XY} (x,y)is 0 but neither $f_X(x)$ nor $f_Y(y)$ are 0.

So, we immediately come to a conclusion that in this case x and y are not independent. And that is intuitively obvious also. Or we can but we can have several experiments for where you have x and y are independent random variables. Say in practice, for example, let say you are measuring the time of birth of some baby and the height of that baby, are these two variables independent? If these two are independent, you can ask the question what is the probability that there was a baby which was born at 11:59pm and with a height of 40 centimeters.

You can ask this question. Your answer would be you would just compute the probability of these two things independently and multiply that activity and that is exactly what this logic here says, F_{XY}(x,y)is simply a $f_X(x) \times f_Y(y)$ if x and y are independent random variables.